

Operational Readiness in Tactical Populations: A Claim-Locked Evidence Synthesis With Bounded Brazilian Public-Safety Adaptation

Running title: Claim-Locked Tactical Readiness Synthesis

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Abstract

Objectives: To integrate heterogeneous tactical-readiness evidence without exceeding the inferential limits of individual studies and to use Brazilian public-safety research as a bounded adaptation case. **Methods:** We conducted an AI-assisted, claim-locked narrative evidence synthesis based on CORE_EVIDENCE_FROZEN (CEF-v1): 9 Core, 1 Supporting, 4 Contextual, and 1 integrity-held record. Evidence roles, prohibited inferences, cohort overlap, and fail-closed integrity controls were preserved. **Results:** Direct evidence covered shift-related nutrition and vigilance, firearm-task outcomes, operational tourniquet use, policing-policy evaluation, a null school-safety program result, academy fitness change, and medical readiness. The largest medical cohort included 7,968 police/fire personnel; the strongest policy design used staggered difference-in-differences across 2,355 municipalities. Tactical-medicine patient outcomes and formal implementation science remained structural gaps. **Conclusions:** Readiness is supported as a multidomain organizational and occupational construct, but the available evidence does not validate a universal score or predictive tool. The proposed framework organizes domains, evidence roles, and transferability boundaries; it remains unvalidated and requires prospective validation before individual, clinical, or operational use.

Keywords: occupational health; operational readiness; tactical athlete; law enforcement; tactical medicine; evidence synthesis

Key Points

- **What is already known:** Tactical readiness is multidimensional and task-linked; established military and cardiovascular frameworks already integrate health, mission demands, and organizational consequences.
- **What this study adds:** A claim-locked architecture separates Core, Supporting, Contextual, and integrity-held evidence, preserves null results and overlapping cohorts, and makes Brazil-to-international transferability explicit.
- **How this may affect research or practice:** Organizations can use the architecture to structure monitoring and research priorities, but the framework must not be used as a validated score, clinical decision tool, or universal readiness threshold.

Introduction

Tactical occupations combine routine occupational exposures with periods of abrupt, mission-critical demand. Law-enforcement officers, firefighters, military personnel, and other tactical professionals may be required to transition

from prolonged vigilance or administrative activity to high-intensity exertion, threat processing, casualty care, armed decision-making, or team-based action with little warning. This combination means that readiness cannot be reduced to aerobic capacity, strength, absence of disease, or technical proficiency in isolation. Total Force Fitness already conceptualized readiness at the unit level as an interaction among demands, resources, and outcomes, while the 2026 ACC/AHA scientific statement on tactical athletes emphasized task-specific cardiovascular demands, environmental and situational strain amplifiers, and the operational consequences of incapacitation.^{1,2} These precedents support a multidimensional view, but they do not resolve how heterogeneous evidence from public-safety systems should be integrated without overstating causality, transferability, or measurement validity.

Brazil provides a useful adaptation case because public safety is decentralized across state institutions, missions differ between police and fire services, training systems are heterogeneous, and available research spans nutrition, shooting, academy fitness, tactical medicine, occupational health, and policy evaluation. The same heterogeneity that makes the evidence informative also creates a high risk of overgeneralization: a small police sample from one state cannot define a national nutrition norm; cadet shooting data cannot be treated as field performance; a state medical cohort cannot estimate national prevalence; and documentation of tourniquet use cannot establish survival benefit. The methodological problem is therefore not simply to summarize studies, but to preserve the inferential boundary of each study while building a coherent operational picture.

The objective of this work was to develop a claim-locked narrative evidence synthesis of operational readiness in tactical populations, using Brazilian public-safety evidence as a bounded adaptation layer. The synthesis was designed to keep null findings visible, prevent double counting of overlapping cohorts, separate central from contextual evidence, and fail closed when integrity was unresolved. Its output is a proposed synthesis framework intended for evidence organization and research planning. It is explicitly not a validated readiness score, diagnostic instrument, fitness-for-duty algorithm, or predictive model.

Conceptual Background

For this synthesis, operational readiness is defined as the state in which an individual and the surrounding organization retain sufficient health, physical capacity, cognitive and technical performance, medical capability, and organizational support to meet mission demands within an acceptable risk envelope. This is a project-level integrative definition, not a replacement for Total Force Fitness or tactical clinical management. The demand-resource logic of Total Force Fitness is relevant because occupational outcomes emerge from interactions between mission demands and available personal or organizational resources.¹ The ACC/AHA tactical-athlete statement adds a complementary clinical perspective by emphasizing that the same task can impose different cardiovascular loads depending on heat, dehydration, chronic alertness, equipment, recovery, and other strain amplifiers.²

These external frameworks establish two important constraints on the present model. First, multidomain readiness is not a novel concept. Second, readiness must remain linked to task context rather than being treated as a single latent trait that can be measured by one score without validation. Accordingly, the synthesis separates eight analytic components: nutrition and schedule load; cognition and firearm-task performance; tactical-medicine practice readiness; organizational and policy readiness; physical and academy readiness; medical readiness; contextual physiological mechanisms; and evidence governance/change control. The first six organize direct or near-direct public-safety evidence, the seventh contains mechanistic or indirectly transferable evidence, and the eighth exists to prevent evidence-role drift.

The framework is therefore deliberately asymmetric. Some domains contain direct Brazilian police data but weak causal designs; others contain stronger quasi-experimental policy designs but outcomes that are organizational rather than physiological; tactical medicine has operational-use documentation but lacks verified comparative patient-outcome evidence; and implementation science remains structurally underdeveloped in the mapped Brazilian literature. This unevenness is not treated as a defect to be hidden. It is treated as information about where the evidence base is mature, where it is only descriptive, and where future validation work is required.

Methods / Evidence-Synthesis Methodology

This article is an AI-assisted, claim-locked narrative evidence synthesis based on the staged evidence program maintained in the project repository. It is not presented as a registered systematic review, meta-analysis, or dual-human-review process. The frozen source of truth was CORE_EVIDENCE_FROZEN — CEF-v1. At the start of this manuscript phase, CEF-v1 contained nine Core records, one Supporting record, four Contextual records, and one Hold Integrity record. Evidence roles were treated as functional constraints: Core evidence could support central claims only within locked wording; Supporting evidence could inform interpretation but could not be silently promoted to Core; Contextual evidence could clarify mechanisms, policy environment, or institutional background without supporting effectiveness claims; and integrity-held evidence was excluded from claim support.

The claim-locking process preserved eleven Brazilian claims. Each claim was linked to one evidence record and one explicit prohibited inference. For example, nutritional evidence permitted association statements but prohibited dietary causality and national norms; firearm studies permitted cadet- and protocol-specific conclusions but prohibited operational universalization; tourniquet documentation permitted a statement that operational use occurred but prohibited mortality, survival, or training-effectiveness claims; and the Paraná medical cohort permitted state-specific association statements but prohibited national prevalence or causal interpretation. These controls were retained verbatim during manuscript expansion.

Human governance and AI assistance were separated. Matheus Florindo de Deus served as human lead reviewer for the Wave 1 Brazilian claim set and has directly confirmed the four ICMJE authorship criteria, with final approval of this specific manuscript version still pending human review. OpenAI ChatGPT (GPT-5.6 Sol; September 20, 2026) supported evidence organization, methodological audit, claim-drift control, and drafting. AI did not act as an author, did not provide independent human verification, and did not assume responsibility for the interpretation or final manuscript. Substantive changes to evidence role, claim wording, certainty, overlap, integrity, prohibited inference, or canonical scientific metadata require a FREEZE_CHANGE_REQUEST; no such request was used for the present writing phase.

Two additional safeguards were applied. First, EV-1473 and EV-1474 were treated as one partially overlapping Espírito Santo firearm-performance cohort family rather than two independent replications. Second, EV-1379 remained HOLD_INTEGRITY / FAIL_CLOSED and contributed no manuscript claim. A reference lock was used to maintain the sixteen source records used in this manuscript, including external conceptual references, Core and Supporting studies, and Contextual institutional or mechanistic sources.

Figure 1 summarizes the evidence-governance architecture used in this project.

Figure 1. Evidence architecture from identification to pre-submission review

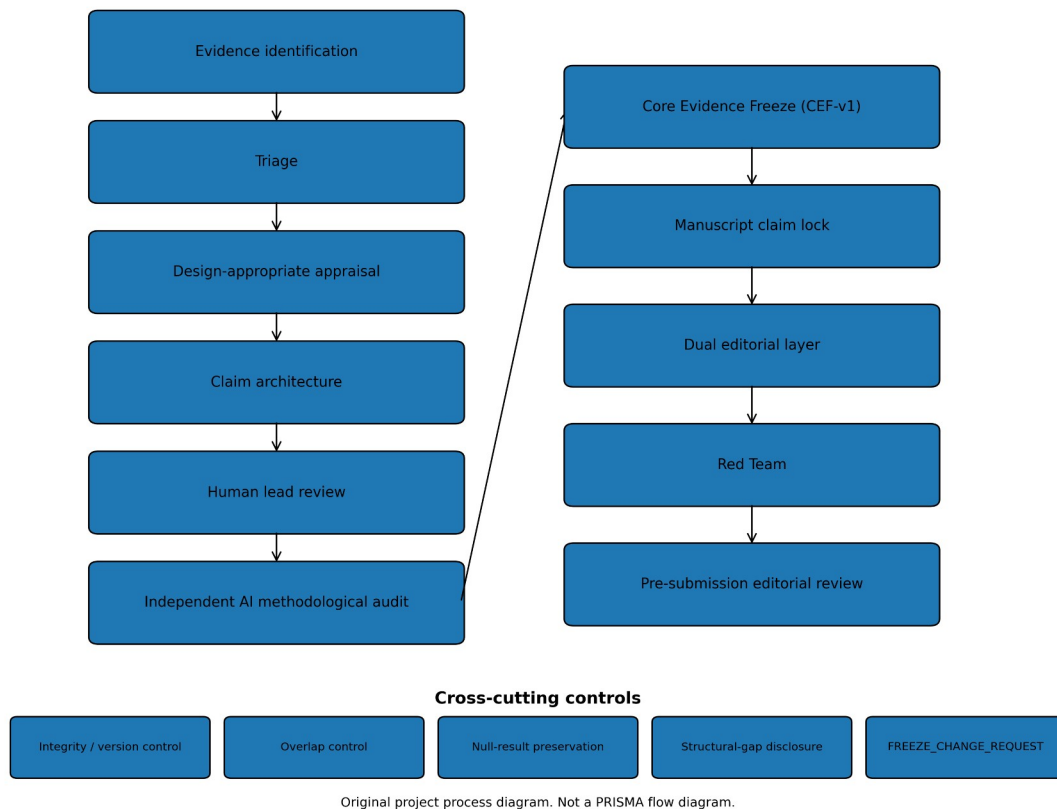


Figure 1. Evidence architecture from identification to pre-submission review. Frozen Batch 10 figure; no scientific modification in v0.12.

Evidence Architecture

The frozen architecture is intentionally heterogeneous. The nine Core records span seven substantive domains: two nutrition/shift-work studies; two firearm-performance publications from a partially overlapping cadet family; one operational tourniquet-use study; two policy/program evaluations; one academy-fitness study; and one medical-readiness cohort.^{3-7,10-12,14} The Supporting record concerns supplemental martial-arts experience and simulated use-of-force performance.¹³ Contextual records include Brazilian APH-Tático material, the national institutional APH-Tático framework, military patrol physiology, and police mental-health promotion.^{8,9,15,16}

This architecture matters because conventional narrative synthesis can inadvertently flatten study design. Here, a cross-sectional association, an uncontrolled longitudinal change, a difference-in-differences policy estimate, a qualitative institutional analysis, and a simulated performance comparison are not treated as equivalent. Instead, the synthesis asks what each design can legitimately contribute to readiness. Cross-sectional work can identify associations and potential monitoring targets; uncontrolled longitudinal work can document change over time but not isolate a causal training component; quasi-experimental policy evaluation can estimate organizational outcomes under identification assumptions but cannot establish physiological readiness; and contextual material can describe institutional structures without being used as proof of effectiveness.

The result is an evidence map rather than an effect-size hierarchy. This is appropriate for a multidomain occupational construct in which outcomes are not commensurable: reaction time, caloric intake, shooting accuracy, tourniquet use, homicide rates, physical-fitness tests, medical readiness, and simulated use-of-force scores cannot be defensibly pooled into one quantitative readiness estimate. The evidence architecture therefore prioritizes traceability, domain-specific interpretation, and explicit non-inference.

Table 1. Frozen Core Evidence Set

Evidence ID	Role	Domain	Design/context	Boundary
EV-1471	CORE	Nutrition/shift work	Local observational police sample	Association only; no national norms
EV-1472	CORE	Chrononutrition/shift work	Regional observational police sample	No causal health/performance inference
EV-1473	CORE	Stress/shooting	PMES cadets	Protocol-specific null; overlap family
EV-1474	CORE	Fitness/shooting	PMES cadets	Cross-sectional; overlap family
EV-1475	CORE	Tactical medicine practice	PMMG records	Use documentation only; no patient outcomes
EV-1478	CORE	Policy/organizational	2,355 municipalities	Policy outcome; not individual readiness
EV-1479	CORE	Program evaluation	Escola Segura	Null result preserved
EV-1481	CORE	Academy fitness	219 male recruits	Uncontrolled pre-post; no component causality
EV-1484	CORE	Medical readiness	Paraná police/fire	State-specific; noncausal

Table 2. Claim-to-evidence map and prohibited inferences

Claim ID	Evidence ID	Permitted claim	Prohibited inference
BR-CLM-001	EV-1471	Association of intake with selected vigilance metrics	No dietary causality / no national norms
BR-CLM-002	EV-1472	Shift-associated intake/meal-timing differences	No national generalization
BR-CLM-003	EV-1473	Most shooting outcomes not significantly worse in tested protocol	No operational-stress universalization
BR-CLM-004	EV-1474	No consistent global fitness-shooting relation; selected associations	No claim fitness is irrelevant
BR-CLM-005	EV-1475	Operational tourniquet use documented	No mortality/survival/effectiveness claim
BR-CLM-006	EV-1478	Estimated -18.3 homicides/100k under DiD assumptions	No physiological-readiness inference
BR-CLM-007	EV-1479	No impact in analyzed Escola Segura outcomes	Do not suppress null result
BR-CLM-008	EV-1480	Better simulated performance with supplemental martial arts experience	No real-world efficacy claim
BR-CLM-009	EV-1481	Fitness improved over six academy months	No causal attribution to components
BR-CLM-010	EV-1483	Military patrol strain as context only	No direct police/fire Core use
BR-CLM-011	EV-1484	Health conditions associated with medical non-readiness in Paraná	No national prevalence/causality

Nutrition and Schedule Load

The nutrition domain is anchored by two observational studies of Brazilian military police working shifts in the northern region. Brito and colleagues studied 22 officers using 24-hour dietary recalls and the Psychomotor Vigilance Test. During day shifts, higher caloric intake correlated with more attention lapses ($r=0.630$) and slower mean reaction time ($r=0.565$); protein intake was also associated with lapses and reaction time at the beginning and end of the shift. During night shifts, psychomotor performance declined, but caloric and macronutrient intake were not

significantly correlated with those performance measures.³ The design is cross-sectional and the sample is small; therefore, the result supports an association between intake patterns and selected vigilance metrics under specific shift conditions, not a claim that higher caloric or protein intake causes impaired police performance.

A second study evaluated 34 military police officers and compared dietary intake and meal timing across day shifts, night shifts, and off-duty days. Median energy intake was higher on day-shift days than on night-shift or off-duty days (2781 versus 2350 and 2295 kcal), with corresponding differences in protein and lipid intake; the eating window also differed by work schedule.⁴ These findings add a chrononutritional dimension to readiness: occupational schedules are associated not only with when officers eat but also with the amount and composition of intake. Yet the evidence does not support a national dietary prescription, nor does it demonstrate that changing meal timing improves operational outcomes.

Taken together, the two studies suggest that schedule and nutrition should be considered jointly when monitoring readiness. The stronger inference is not that a particular macronutrient target optimizes performance, but that shift-dependent eating behavior and vigilance can vary together in operational populations. This distinction is important for intervention design. A future program should prospectively measure sleep, workload, dietary timing, total intake, environmental conditions, and task performance rather than treating nutrition as an isolated exposure. It should also avoid assuming that patterns observed in Acre generalize to different police systems, climates, schedules, or demographic profiles.

Cognition and Firearm-Task Performance

Firearm-task performance is represented by two publications from a partially overlapping Espírito Santo Military Police cadet cohort family. In the stress-symptom study, 24 third-year officer cadets were classified according to the presence of stress symptoms. Shooting was assessed using a standardized defensive-shooting evaluation at 5 m, and handgrip strength was measured in dominant, non-dominant, and shooting positions. Shooting time, total score, and the derived accuracy metric did not differ significantly between groups. Handgrip strength in the shooting position was lower in cadets with stress symptoms, whereas dominant and non-dominant handgrip measures were not significantly different.⁵ The bounded conclusion is therefore that the tested stress-symptom classification was associated with a position-specific grip-strength difference but not with a detectable decrement in the measured shooting outcomes under that protocol.

The second publication assessed 26 military police officers and examined relationships among institutional physical-fitness indicators, handgrip strength, mood state, body-mass-related variables, and firearm proficiency. Overall physical-fitness indicators did not show a consistent global correlation with shooting proficiency; selected associations were observed for dominant handgrip strength, body-mass-related variables, and mood vigor.⁶ This does not mean that physical fitness is irrelevant to armed policing. The shooting task samples a narrow performance domain, and fitness may affect mobility, fatigue tolerance, casualty movement, use of force, heat strain, or the capacity to repeat high-intensity tasks without being strongly correlated with a short firearm test.

The cohort-overlap rule is methodologically important. The two publications answer different questions, but they cannot be counted as two independent replications that a broad category such as 'fitness' does not affect shooting. The appropriate synthesis is narrower: within the studied cadet/officer samples and protocols, gross physical-fitness measures were not consistently aligned with firearm score, while specific psychophysiological or strength variables showed selected relationships. Operational validation would require independent cohorts, ecologically valid scenarios, repeated-measures designs, and performance outcomes that capture decision quality, accuracy, time, movement, threat discrimination, and safety under fatigue or stress.

Tactical Medicine Readiness

The tactical-medicine domain illustrates the difference between operational documentation and clinical effectiveness. Matos and Santos combined documentary/normative analysis with PMMG occurrence records from 2013-2017 and 2020-2024. The study documented tourniquet use by military police in third parties, with civilian victims representing more than 90% of recorded applications, and described increased use after the institutionalization of training.⁷ For

this synthesis, those records establish that tourniquets were operationally used and that the practice was institutionally relevant. They do not establish survival benefit, mortality reduction, comparative effectiveness, appropriateness in every case, or the causal effect of training.

Contextual sources reinforce that APH-Tático has an institutional and doctrinal presence in Brazilian public safety. A Paraná-focused qualitative review describes tactical prehospital care in special-operations settings, while the Ministry of Justice and Public Security maintains the national APH-Tático directive, minimum competencies, equipment guidance, and curriculum materials.^{8,9} These sources help define practice context and governance but were not promoted to effectiveness evidence. Their role is particularly important because a mature readiness system requires not only individual medical skills but also equipment availability, protocols, training continuity, scene integration, data capture, and interfaces with the civilian emergency-care system.

The central limitation remains explicit: **STRUCTURAL_EVIDENCE_GAP — BRAZILIAN PATIENT-OUTCOME TACTICAL MEDICINE**. The frozen evidence set contains operational-use documentation but no robust verified comparative patient-outcome study demonstrating the effect of tactical-medicine implementation on survival, hemorrhage control, disability, time to definitive care, or preventable death in Brazilian public-safety operations. Closing this gap requires interoperable registries linking scene variables, interventions, provider role, injury characteristics, transport, hospital outcomes, and training exposure. Until such data exist, tactical-medicine readiness should be discussed as capability and documented practice, not as proven patient-outcome effectiveness.

Organizational and Policy Readiness

Organizational readiness is supported by policy evidence with substantially different designs. Parfitt and colleagues evaluated results-based, data-driven policing reforms implemented across nine Brazilian states. Using a staggered difference-in-differences design over 2,355 municipalities and a long municipal panel, the study estimated an average reduction of 18.3 homicides per 100,000 after implementation, with stronger effects in higher-violence settings and vulnerable groups.¹⁰ This is the strongest quasi-experimental signal in the frozen Brazilian set, but its outcome is public-safety performance at the municipal level. It should not be reinterpreted as proof of individual officer fitness, health, or tactical proficiency.

The Escola Segura evaluation provides an equally important counterweight. The program placed retired or reserve military police in state schools, and its impact on perceived safety, victimization, graffiti, and illicit-drug use was estimated using a quasi-experimental difference-in-differences comparison between a treated school and a similar untreated school. The study reported no impact on the analyzed outcomes.¹¹ Preserving this null result is central to the evidence-governance model: readiness-oriented policy should not assume that a plausible or visible intervention is effective merely because it has institutional support.

Together, these studies show why organizational readiness must include evaluation capacity. Data-driven management can be associated with large public-safety effects under a credible identification strategy, while another security program can show no measurable impact on selected outcomes. This contrast argues for outcome specification before implementation, transparent analytic plans, and the use of designs capable of separating program effects from secular trends. Yet the mapped literature does not provide a mature body of CFIR-, ERIC-, Proctor-, or RE-AIM-style implementation research in Brazilian public safety. The unresolved label is therefore **STRUCTURAL_METHODS_GAP — FORMAL IMPLEMENTATION SCIENCE IN BRAZILIAN PUBLIC SAFETY**.

Physical and Academy Readiness

Physical readiness is represented by an observational longitudinal study of 219 male military police recruits in Tocantins. Health-related physical fitness was measured before and after six months of mandatory academy training, with an earlier assessment after a seven-month unsupervised period. The recruits maintained high fitness before academy entry and improved across all measured fitness components during the academy; reported average changes ranged from 7.7% to 69.0%, while cardiorespiratory fitness improved by less than 10%.¹² The study demonstrates that measurable fitness changed over an academy period, but because there was no external control group and the

academy included multiple simultaneous exposures, the data cannot isolate which training components caused the improvements.

This distinction matters operationally. Academy fitness gains show that readiness is modifiable during structured training, but they do not define the optimal program, the durability of gains after graduation, or the relationship between test improvement and field performance. A readiness system should therefore track both capacity and transfer: aerobic fitness, strength/endurance, and other physical metrics are relevant, but they should be linked prospectively to occupational tasks, injury, fatigue, attendance, recovery, and retention rather than treated as ends in themselves.

The Supporting martial-arts study adds a different perspective. Thirty-four cadets were grouped according to supplemental Brazilian jiu-jitsu, Muay Thai/kickboxing, or no supplemental martial-arts experience and completed a simulated domestic-violence use-of-force call. Trained groups performed better on several simulated outcomes, particularly movement execution, effectiveness, final grade, and collective scores; some mean differences approached one fifth of the respective scale range.¹³ Because exposure was not randomized, the sample was small and convenience-based, and the outcome was a simulation, the study remains Supporting evidence. It suggests a training hypothesis worthy of controlled operational study, not proof that martial arts improves real-world force outcomes or safety.

Medical and Cardiovascular Readiness

Medical readiness is represented most directly by the Paraná cross-sectional cohort of 7,968 professionals: 6,621 police officers and 1,347 firefighters who underwent periodic health assessments. Police officers had less favorable health profiles than firefighters on several measures, and musculoskeletal disorders were the most commonly reported disease group. Among men, hypertension, psychiatric disease, and dyslipidemia were prominent; among women, psychiatric disease was the second most prevalent category. Male police officers aged 40 years or older had the highest estimated probability of not being medically ready for duty (40.1%). Diabetes, dyslipidemia, hypertension, high total cholesterol, and heart disease were each associated with higher odds of medical non-readiness.¹⁴

These data are important because they connect chronic health conditions with an occupational disposition—medical readiness—rather than merely describing disease prevalence. At the same time, the study is cross-sectional and state-specific. The associations do not establish that the identified conditions caused non-readiness, and the Paraná estimates cannot be presented as national prevalence. Differences between police and firefighters may reflect age, occupational selection, work organization, exposures, health-care access, or other measured and unmeasured factors.

The cardiovascular interpretation is strengthened conceptually, not quantitatively, by the ACC/AHA tactical-athlete statement. That guidance emphasizes that fit-for-full-duty decisions require attention to task demands, team and mission consequences, and cardiovascular strain amplifiers such as heat, dehydration, suboptimal recovery, and high alert.² A military contextual study illustrates one such mechanism: 20 Brazilian peacekeepers completing an approximately 161-minute patrol in Haiti expended about 711 kcal on average and showed body-mass loss and autonomic shifts; changes in sympathovagal balance were correlated with energy expenditure, aerobic fitness, and body-mass loss.¹⁵ Because those participants were soldiers in Haiti, the study remains contextual for Brazilian police and firefighters rather than Core evidence. It supports plausibility that environment, hydration, and mission load can amplify physiological strain, but it does not quantify police readiness thresholds.

Integrated Operational Readiness Interpretation

Across domains, the strongest conclusion is not a numerical readiness estimate but a pattern of interdependence. Schedules shape eating behavior; nutrition and vigilance can vary together; physical capacity can improve during training without necessarily correlating with a narrow firearm score; chronic health conditions can be associated with duty readiness; organizational reforms can affect community-level outcomes; and tactical medical capability can exist without patient-outcome validation. These findings are heterogeneous by design, but they converge on one operational principle: readiness is produced by the interaction of person, task, environment, training, health system, and organization.

The evidence also shows why a single composite score would be premature. Domains differ in measurement level—from individual reaction time to municipality-level homicide rates—and in causal strength. Converting them into a single index would require explicit construct definition, item selection, scaling, weighting, reliability testing, measurement invariance, criterion validation, and prospective outcome validation. None of those psychometric or predictive steps has yet been completed. Consequently, the present framework is useful as an organizing ontology and governance scaffold, but not as a score.

A second integrative lesson concerns null and boundary information. The absence of significant shooting differences by stress-symptom group under one protocol, the lack of a global fitness-shooting correlation, and the null Escola Segura result are not failures of the literature; they prevent the framework from becoming selectively positive. Similarly, the tourniquet and martial-arts studies are informative because their limitations are retained. An evidence system that preserves negative, null, indirect, and noncausal findings is more useful for decision-making than one that accumulates only supportive signals.

Proposed Operational-Readiness Framework

The project-proposed framework contains eight linked components. (1) Nutrition/schedule load captures work timing, eating window, intake patterns, sleep-recovery interfaces, and their relationship to vigilance. (2) Cognition/firearm task captures attention, stress symptoms, mood, grip characteristics, decision demands, and technical firearm performance. (3) Tactical-medicine practice readiness includes individual skills, equipment, protocols, training, documentation, scene integration, and links to patient outcomes. (4) Organizational/policy readiness captures management systems, data use, accountability, program design, and evaluation capability. (5) Physical/academy readiness encompasses fitness capacities, structured training, task transfer, and durability of adaptation. (6) Medical readiness covers chronic disease, injury, cardiovascular risk, periodic assessment, and fit-for-duty status. (7) Contextual physiological mechanisms include heat, hydration, autonomic load, equipment, and mission strain when direct population evidence is unavailable. (8) Evidence-governance/change control defines what evidence may support a claim and how updates are adjudicated.

Relationships among components are hypothesized rather than quantified. Shift schedules may influence recovery and eating behavior; health and fitness may condition tolerance to mission load; cognitive state and technical skill jointly affect task performance; tactical-medicine capability may mediate consequences of injury; and organizational systems determine whether training, health surveillance, equipment, and data are sustained. These pathways are scientifically plausible and consistent with the mapped evidence, but the framework has not established causal pathways, weights, thresholds, or predictive accuracy.

A future validation program should therefore proceed in stages. Content validity should first be established with multidisciplinary experts and operational stakeholders. Candidate measures should then be specified for each domain, followed by reliability and feasibility testing across police, fire, military, and emergency-service settings. Construct validity should test expected relationships without assuming that all domains collapse into one factor. Criterion and predictive validation should prospectively examine meaningful outcomes such as task performance, injury, medical restriction, training attrition, clinical events, or mission-relevant endpoints. External validation should assess performance across jurisdictions, sexes, age groups, specialties, climates, and duty schedules. Until such work is completed, the label remains: PROPOSED SYNTHESIS FRAMEWORK — NOT YET VALIDATED.

Figure 2. Proposed operational-readiness synthesis framework

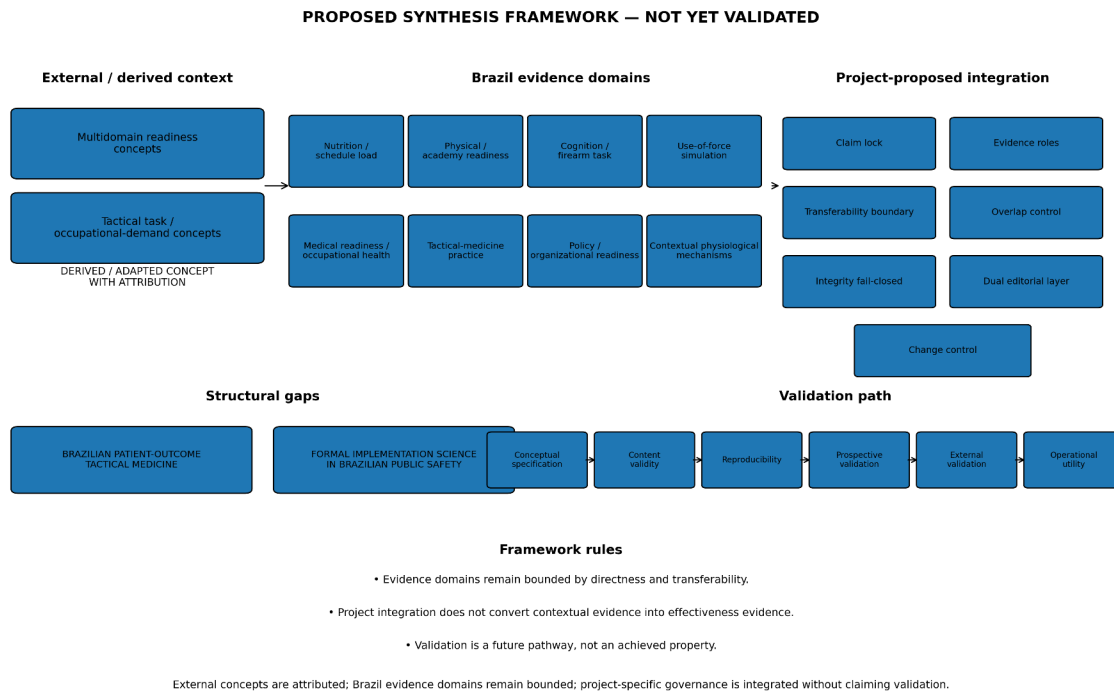


Figure 2. Proposed synthesis framework and future validation pathway. PROPOSED SYNTHESIS FRAMEWORK — NOT YET VALIDATED.

International Applicability

The Brazilian evidence should be used internationally as a bounded adaptation case. Some mechanisms and governance principles are likely portable: shift work can alter eating behavior; operational tasks can interact with cardiovascular strain; organizational reforms should be evaluated; and readiness programs benefit from transparent evidence roles. However, effect estimates are not automatically portable. Police academy structures, firearms training, medical standards, labor arrangements, tactical-medicine scope, crime patterns, and public-health systems differ across jurisdictions.

The appropriate international transfer process is therefore domain-by-domain. A foreign organization could adopt the evidence-governance method—Core/Supporting/Contextual/Hold roles, claim locks, prohibited inferences, overlap control, and change requests—without assuming that Brazilian numerical results apply locally. It could then populate the architecture with local evidence and compare whether the same structural gaps remain. This makes the framework a template for disciplined synthesis rather than a Brazilian score exported abroad.

Implementation Considerations

Implementation should begin with data architecture rather than a new composite metric. Organizations can identify existing data streams for schedules, training, fitness, periodic medical assessment, injury, tactical-medicine use, and operational outcomes, then define which questions each stream can legitimately answer. The evidence-role model can be used operationally: direct high-quality local evidence supports local decisions; indirect or simulated evidence generates hypotheses; contextual evidence informs mechanisms; and integrity concerns block use until resolved.

Several implementation safeguards follow directly from the frozen claims. Tourniquet-use counts should not be reported as lives saved unless linked patient outcomes support that statement.⁷ Municipal policy effects should not be presented as individual officer readiness.¹⁰ Simulated use-of-force performance should not be treated as field effectiveness.¹³ Military heat/hydration physiology should not be converted into police thresholds without population-

specific validation.¹⁵ The mental-health promotion literature can inform organizational context and service design, but qualitative program development should not be treated as evidence of clinical effectiveness.¹⁶

A practical implementation pathway would combine a minimum dataset, periodic multidisciplinary review, and prospective evaluation. The minimum dataset should be small enough to sustain in routine operations but rich enough to distinguish exposure, capacity, response, and outcome. Crucially, implementation studies should predefine feasibility, acceptability, adoption, fidelity, reach, and sustainability alongside performance or health outcomes. That requirement directly addresses the current structural methods gap in Brazilian public safety.

Discussion

This synthesis shows a field with meaningful signals but uneven evidentiary depth. Nutrition and schedule research provides direct occupational data, yet sample sizes are small and designs are observational. Firearm studies provide task-specific measurement but are concentrated in cadets and a partially overlapping cohort family. Academy fitness shows longitudinal improvement but lacks a control capable of isolating causal components. Medical readiness has the largest direct occupational cohort, but remains cross-sectional and geographically specific. Policy evaluation includes a comparatively strong staggered difference-in-differences design, although its outcome operates at the municipal rather than individual level. Tactical medicine has institutional maturity and documented use but lacks verified patient-outcome effectiveness evidence. This pattern argues against both nihilism and overconfidence: there is enough evidence to organize readiness scientifically, but not enough to validate a universal readiness instrument.

The domains also differ in what transfer means. A physiological association may transfer mechanistically but not numerically; a policy design may transfer methodologically but depend on local institutions; a tactical-medicine protocol may transfer doctrinally while patient outcomes depend on trauma patterns, transport, scope of practice, and system integration. The framework therefore treats transferability as an explicit judgment rather than an implicit assumption. This is especially important in international tactical-athlete literature, where military, fire, police, and emergency medical populations are often discussed together despite important differences in mission and governance.

The integration with established frameworks is deliberately limited. Total Force Fitness contributes a demand-resource-outcome logic, and the ACC/AHA statement contributes task-specific cardiovascular reasoning and strain amplifiers.^{1,2} The present work does not rename those concepts as original. Its new element is the combination of domain organization with claim locking, evidence-role separation, structural-gap labeling, overlap control, and a Brazil-to-international adaptation layer. That combination may improve transparency in multidisciplinary readiness research because it makes the permissible scope of each claim auditable.

The null and noncausal evidence is particularly instructive. A readiness program built only from positive findings could mistakenly recommend interventions unsupported by the actual evidence. Here, the Escola Segura null result remains visible; the firearm studies do not support a broad claim that stress or physical fitness has no operational relevance; and the tourniquet study does not become a mortality study.^{5-7,11} These restrictions are scientifically conservative, but they increase usefulness by narrowing the distance between what the source showed and what the synthesis says.

Two structural gaps should define the next research cycle. First, tactical medicine needs patient-level outcomes linked across the prehospital and hospital continuum. Second, public-safety interventions need formal implementation research so that effectiveness, feasibility, fidelity, and sustainability are measured together. The absence of these data currently prevents the framework from supporting outcome-based tactical-medicine recommendations or mature implementation guidance.

Operational Implications

For practitioners and organizations, the framework supports a portfolio approach. Readiness surveillance should not be limited to annual fitness testing or medical clearance; it should consider schedule load, recovery, task performance, medical risk, tactical-medicine capability, and organizational conditions. However, surveillance should be proportionate and ethically governed. Measures should have a defined purpose, clear ownership, privacy protections,

and a prespecified response when risk is detected. Data collection without an actionable pathway can create burden without improving readiness.

Training programs should distinguish proximal from distal outcomes. Academy fitness tests are proximal capacity measures; simulated use-of-force scores are performance proxies; firearm scores are task measures; and field injuries, restrictions, or community outcomes are distal outcomes. Interventions should not be judged only on the measure they directly train. A strong readiness program links training exposure to a chain of outcomes while preserving causal uncertainty.

Finally, leaders should treat evidence governance as part of readiness. The ability to identify retractions or integrity concerns, prevent double counting, preserve null findings, and document changes protects operational decisions from evidence drift. In this project, EV-1379 remained fail-closed and was not allowed to support claims. That governance behavior is itself an organizational capability relevant to evidence-informed public safety.

Limitations

Several limitations constrain this synthesis. It is a narrative, claim-locked synthesis rather than a registered systematic review or meta-analysis, and study identification and adjudication were embedded in a staged project workflow. The human governance model included a lead human reviewer and an independent AI methodological audit, not two independent human reviewers. Consequently, the process should not be described as conventional dual screening or dual extraction.

The empirical base is small and heterogeneous. Several Core studies have small or geographically restricted samples; firearm evidence is concentrated in cadets and includes a partially overlapping cohort family; academy fitness is uncontrolled; simulated use-of-force evidence is nonrandomized; medical readiness is cross-sectional and from one state; and military patrol physiology is indirect for police and fire services. The policy studies address organizational outcomes rather than individual readiness. These design differences limit synthesis across domains and make quantitative pooling inappropriate.

The framework itself has not undergone content-validity panels, psychometric testing, external validation, threshold derivation, or predictive evaluation. It must therefore not be used for individual clinical decisions, employment decisions, fitness-for-duty classification, or automated risk scoring. EV-1379 remains HOLD_INTEGRITY / FAIL_CLOSED and was excluded from claim support. Two structural gaps also remain unresolved: Brazilian patient-outcome tactical medicine and formal implementation science in Brazilian public safety.

Future Research Agenda

Research should progress from isolated studies toward linked prospective systems. Priority one is a multi-institutional tactical-medicine registry that records mechanism of injury, scene conditions, responder profile, interventions, time intervals, transport, hospital linkage, and patient outcomes. Priority two is multi-state medical-readiness surveillance with harmonized variables and transparent denominators. Priority three is independent operational firearm and decision-performance research outside academy samples, with repeated measures under fatigue, time pressure, and ecologically valid stressors. Priority four is prospective nutrition/schedule research integrating sleep, chronotype, heat, workload, dietary timing, and vigilance rather than analyzing these domains separately.

Training studies should use designs capable of isolating causal components when feasible, including randomized or cluster-randomized approaches, stepped-wedge designs, controlled cohorts, or strong quasi-experiments. Use-of-force research should move from simulation to carefully governed field-proximal outcomes without sacrificing ethics or safety. Implementation studies should measure not only whether a program changes outcomes but also whether it is adopted, delivered with fidelity, accepted by personnel, equitable across subgroups, and sustainable.

Framework validation should be a distinct program of research rather than an assumed consequence of publication. The proposed components should be independently reviewed, operationalized, tested for reliability and feasibility, and then evaluated against prespecified outcomes. Only after external validation should thresholding, weighting, or predictive use be considered.

Table 3. Structural evidence gaps and research priorities

Gap	Current limitation	Research priority
Brazilian patient-outcome tactical medicine	No robust comparative patient-outcome effectiveness evidence verified	Develop interoperable APH-Tático registries with patient outcomes
Formal implementation science in Brazilian public safety	No verified CFIR/ERIC/Proctor/RE-AIM-style evaluation in mapped Brazil public-safety set	Prospective hybrid/implementation studies in public-safety settings

Conclusion

Operational readiness in tactical populations is multidimensional, context dependent, and methodologically vulnerable to overgeneralization. The frozen evidence set supports bounded conclusions on shift-related nutrition, vigilance, firearm-task performance, tactical-medicine practice, academy fitness, medical readiness, use-of-force simulation, and organizational policy. It does not support a universal readiness score or a single causal model. The principal contribution of this synthesis is a transparent architecture that links domain evidence to explicit claim boundaries and adapts Brazilian public-safety findings without converting local estimates into universal parameters. The resulting model remains a PROPOSED SYNTHESIS FRAMEWORK — NOT YET VALIDATED. Its value at this stage is to organize evidence, expose structural gaps, and define a testable validation agenda rather than to classify individuals or predict operational success.

Declarations

Acknowledgments: [HUMAN CONFIRMATION REQUIRED].

Author Contributions: Matheus Florindo de Deus has confirmed all four ICMJE authorship criteria. Final CRediT roles remain [HUMAN CONFIRMATION REQUIRED] before submission. Additional authors, if any, require individual confirmation.

Conflict of Interest: [HUMAN CONFIRMATION REQUIRED] by all final authors.

Data Availability: Evidence-governance and audit artifacts are maintained in the project repository and associated archive. Public-access status at submission: [HUMAN CONFIRMATION REQUIRED].

Funding: [HUMAN CONFIRMATION REQUIRED].

AI Use Disclosure: OpenAI ChatGPT (GPT-5.6 Sol; September 20, 2026) supported evidence organization, methodological audit, claim-drift control, drafting/editing, and audit-matrix generation. AI was not an author, did not assume scientific responsibility, and did not substitute for human accountability or final approval.

Access and adjudication limitations added in v0.13

This v0.13 revision incorporates a reference-saturation partial pass rather than a final scientific pass. Full text adequate for appraisal remained unavailable or insufficient for 22 HIGH records, five targeted MEDIUM records, and 15 PMC100 records. The fail-closed rule means that none of those records is assumed to support the present interpretation.

Three HIGH records with potential contradictory implications remained unadjudicated because lawful full text was unavailable: EV-0052 for psychological self-report as a performance-screening surrogate, EV-0140 for short-term moderately high-fat dietary effects on performance, and EV-1066 for BMI as a global proxy of operational readiness. Accordingly, this manuscript does not make definitive psychological self-report screening claims, does not make universal dietary-fat claims, and does not use BMI alone as a readiness indicator.

Provisional candidates are not final evidence. EV-1462 may only contextualize task- and context-dependent partial mitigation of selected cognitive or operational consequences of sleep loss by caffeine; it does not establish that caffeine reverses or solves sleep loss. EV-1463 is limited to cross-sectional cardiometabolic observations in one

PMDF cohort, and EV-1466 is exploratory context only (AUC approximately 0.58). Named human review is required before any candidate is promoted to a final scientific claim or reference decision.

The framework status remains exactly: PROPOSED SYNTHESIS FRAMEWORK — NOT YET VALIDATED. CEF-v1 is unchanged.

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Human Final Review Record

Reviewer: Matheus Florindo de Deus

Status: PENDING_HUMAN_FINAL_REVIEW

Decision after full reading: ☐ APPROVE ☐ APPROVE WITH CORRECTIONS ☐ MAJOR REVISION
REQUIRED

Comments: